



COMP3000

Programming Languages

Session 2, In person-scheduled-weekday, North Ryde 2026

School of Computing

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General Information

Unit convenor and teaching staff

Unit Convenor

Lachlan Patrick

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Lecturer

Matthew Roberts

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Credit points

10

Prerequisites

130cp at 1000 level or above including COMP2010 or COMP2000

Corequisites

Co-badged status

Unit description

Formal languages play a central role in modern software development. Programming languages such as Java and C++ allow developers to express their algorithms and data structures. Compilers and interpreters transform programs into running software. This unit studies software languages by looking at how they are used in software development. Students will study how to formally understand the syntax, semantics and translation of software languages. Practical exercises involve writing software language processors of various kinds such as simple compilers or data transformation tools.

Important Academic Dates

Information about important academic dates including deadlines for withdrawing from units are available at <https://www.mq.edu.au/study/calendar-of-dates>

Learning Outcomes

On successful completion of this unit, you will be able to:

ULO1: evaluate the role that languages play in software development and describe a spectrum of software languages that are in current use

ULO2: express properties of software languages using formal notations

ULO3: translate formal notations of software language properties into implementations of

language processors

ULO4: defend the correct operation of a language processor by construction and use of appropriate test cases

General Assessment Information

Release

Specifications for the assignments will be released no later than

- Assignment 1 - 7th September.
- Assignment 2 - 19th October.

Requirements to Pass this Unit

To pass this unit you must achieve a total mark equal to or greater than 50%. There are no hurdle requirements for the unit.

LATE ASSESSMENT SUBMISSION PENALTY

Unless a Special Consideration request has been submitted and approved, a 5% penalty (of the total possible mark of the task) will be applied for each day a written report or presentation assessment is not submitted, up until the 7th day (including weekends). After the 7th day, a grade of '0' will be awarded even if the assessment is submitted. The submission time for all uploaded assessments is **11:55 pm**. A 1-hour grace period will be provided to students who experience a technical concern. For any late submission of time-sensitive tasks, such as scheduled tests/exams, performance assessments/presentations, and/or scheduled practical assessments/labs, please apply for [Special Consideration](#). For example, if the assignment is worth 25 marks (of the entire unit) and your submission is late by 19 hours (or 23 hours 59 minutes 59 seconds), 1.25 marks (5% of 25 marks) will be deducted. If your submission is late by 24 hours (or 47 hours 59 minutes 59 seconds), 2.5 marks (10% of 25 marks) will be deducted, and so on.

ASSESSMENTS WHERE LATE SUBMISSIONS WILL BE ACCEPTED

- Assignment 1: YES, Standard Late Penalty applies
- Assignment 2: YES, Standard Late Penalty applies
- Exam: NO

Special Consideration

The Special Consideration Policy aims to support students who have been impacted by short-term circumstances or events that are serious, unavoidable and significantly disruptive, and which may affect their performance in assessment.

If you experience circumstances or events that affect your ability to complete either of the assignments in this unit on time, please inform the convenor and submit a Special Consideration request through ask.mq.edu.au.

Generative AI Usage

Assessments 1 (Programming Task) and 2 (Programming Task) are AI Open while Assessment 3 (Final Exam) is AI Observed.

The university has taken a 'two lane' approach with GenAI: assessments can be AI Open (meaning students complete the assessment in their own time and may use GenAI) and Observed (students complete assessments in supervised environments, where use of AI is controlled or not permitted). More information on the use of GenAI in Assessment can be found [here](#).

Assignment One

Assessment Type ¹: Programming Task

Indicative Time on Task ²: 25 hours

Due: **27/09/2026**

Weighting: **25%**

Groupwork/Individual: Individual

Short extension ³: No

AI Approach: Open

A programming task to demonstrate practical mastery of the first half of the course.

On successful completion you will be able to:

- express properties of software languages using formal notations
- translate formal notations of software language properties into implementations of language processors
- defend the correct operation of a language processor by construction and use of appropriate test cases

Assignment Two

Assessment Type ¹: Programming Task

Indicative Time on Task ²: 25 hours

Due: **08/11/2026**

Weighting: **25%**

Groupwork/Individual: Individual

Short extension ³: No

AI Approach: Open

A programming task to demonstrate mastery of the second half of the course.

On successful completion you will be able to:

- express properties of software languages using formal notations,
- translate formal notations of software language properties into implementations of language processors,
- defend the correct operation of a language processor by construction and use of appropriate test cases.

Examination

Assessment Type ¹: Examination

Indicative Time on Task ²: 25 hours

Due: **During the formal exam period**

Weighting: **50%**

Groupwork/Individual: Individual

Short extension ³: No

AI Approach: Closed

An exam to test both theoretical and practical mastery of all the course material

On successful completion you will be able to:

- evaluate the role that languages play in software development and describe a spectrum of software languages that are in current use,
- express properties of software languages using formal notations,
- translate formal notations of software language properties into implementations of language processors,
- defend the correct operation of a language processor by construction and use of appropriate test cases.

¹ If you need help with your assignment, please contact:

- the academic teaching staff in your unit for guidance in understanding or completing this type of assessment
- Academic Success for academic skills support.

² Indicative time-on-task is an estimate of the time required for completion of the assessment task and is subject to individual variation.

³ An automatic short extension is available for some assessments. Apply through the Service Connect Portal.

Assessment Tasks

Name	Weighting	Hurdle	Due	Groupwork/Individual	Short Extension	AI Approach
<u>Final Exam</u>	50%	No	During the formal exam period	Individual	No	Observed
<u>Assignment One</u>	25%	No	27/09/2026	Individual	No	Open
<u>Assignment Two</u>	25%	No	08/11/2026	Individual	No	Open

Final Exam

Assessment Type ¹: Examination

Indicative Time on Task ²: 25 hours

Due: **During the formal exam period**

Weighting: **50%**

Groupwork/Individual: Individual

Short extension ³: No

AI Approach: Observed

You will undertake a final examination during the formal examination period.

On successful completion you will be able to:

- evaluate the role that languages play in software development and describe a spectrum of software languages that are in current use
- express properties of software languages using formal notations
- translate formal notations of software language properties into implementations of language processors
- defend the correct operation of a language processor by construction and use of appropriate test cases

Assignment One

Assessment Type ¹: Experiential task

Indicative Time on Task ²: 25 hours

Due: **27/09/2026**

Weighting: **25%**

Groupwork/Individual: Individual

Short extension ³: No

AI Approach: Open

A programming task to demonstrate practical mastery of the first half of the course

On successful completion you will be able to:

- express properties of software languages using formal notations
- translate formal notations of software language properties into implementations of language processors
- defend the correct operation of a language processor by construction and use of appropriate test cases

Assignment Two

Assessment Type ¹: Experiential task

Indicative Time on Task ²: 25 hours

Due: **08/11/2026**

Weighting: **25%**

Groupwork/Individual: Individual

Short extension ³: No

AI Approach: Open

A programming task to demonstrate mastery of the second half of the course.

On successful completion you will be able to:

- express properties of software languages using formal notations
- translate formal notations of software language properties into implementations of language processors
- defend the correct operation of a language processor by construction and use of appropriate test cases

¹ If you need help with your assignment, please contact:

- the academic teaching staff in your unit for guidance in understanding or completing this type of assessment
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Delivery and Resources

COMP3000 is taught via weekly 2-hour lectures and practical classes.

Each week you should participate in your scheduled two hour practical. For details of scheduled classes consult the [timetables webpage](#).

Week1

Note that classes commence in **week 1**. The week-by-week details of the practical classes will be available from iLearn.

Students are expected to do significant preparatory work, readings and exercises *before* attending classes.

Textbook and Reading Materials

The required text for the unit is [Crafting Interpreters by Robert Nystrom](#)

Each week a set of notes, a Rapid Aptitude Test (RAT), an application exercise, and self-study material will be provided in a pdf file. This material will include a set of activities (including the RAT) which are to be done before class.

Communication Methods in COMP3000

We will communicate with you via your university email and through announcements on iLearn. Queries to convenors can either be placed on the iLearn discussion board or sent to the unit convenor via the contact email on iLearn.

Policies and Procedures

Macquarie University policies and procedures are accessible from [Policy Central](#) (<https://policies.mq.edu.au>). Students should be aware of the following policies in particular with regard to Learning and Teaching:

- [Academic Appeals Policy](#)
- [Academic Integrity Policy](#)
- [Academic Progression Policy](#)
- [Assessment Policy](#)
- [Fitness to Practice Procedure](#)
- [Assessment Procedure](#)
- [Complaints Resolution Procedure for Students and Members of the Public](#)
- [Special Consideration Policy](#)

Students seeking more policy resources can visit [Student Policies](#) (<https://students.mq.edu.au/support/study/policies>). It is your one-stop-shop for the key policies you need to know about throughout your undergraduate student journey.

To find other policies relating to Teaching and Learning, visit [Policy Central \(https://policies.mq.edu.au\)](https://policies.mq.edu.au) and use the [search tool](#).

Student Code of Conduct

Macquarie University students have a responsibility to be familiar with the Student Code of Conduct: <https://students.mq.edu.au/admin/other-resources/student-conduct>

Results

Results published on platform other than [eStudent](#), (eg. iLearn, Coursera etc.) or released directly by your Unit Convenor, are not confirmed as they are subject to final approval by the University. Once approved, final results will be sent to your student email address and will be made available in [eStudent](#). For more information visit connect.mq.edu.au or if you are a Global MBA student contact globalmba.support@mq.edu.au

Academic Integrity

At Macquarie, we believe [academic integrity](#) – honesty, respect, trust, responsibility, fairness and courage – is at the core of learning, teaching and research. We recognise that meeting the expectations required to complete your assessments can be challenging. So, we offer you a range of resources and services to help you reach your potential, including free [online writing and maths support](#), [academic skills development](#) and [wellbeing consultations](#).

Student Support

Macquarie University provides a range of support services for students. For details, visit <http://students.mq.edu.au/support/>

Academic Success

[Academic Success](#) provides resources to develop your English language proficiency, academic writing, and communication skills.

- [Ask a Study Coach](#)
- [Access StudyWISE](#)
- [Upload an assignment to Studiosity](#)
- [Complete the Academic Integrity Module](#)

The Library provides online and face to face support to help you find and use relevant information resources.

- [Subject and Research Guides](#)
- [Ask a Librarian](#)

Student Services and Support

Macquarie University offers a range of [Student Support Services](#) including:

- [IT Support](#)

- [Accessibility and disability support](#) with study
- Mental health [support](#)
- [Safety support](#) to respond to bullying, harassment, sexual harassment and sexual assault
- [Social support including information about finances, tenancy and legal issues](#)
- [Student Advocacy](#) provides independent advice on MQ policies, procedures, and processes

Student Enquiries

Got a question? Ask us via the [Service Connect Portal](#), or contact [Service Connect](#).

IT Help

For help with University computer systems and technology, visit <https://students.mq.edu.au/support/technology/service-desk>.

When using the University's IT, you must adhere to the [Acceptable Use of IT Resources Policy](#). The policy applies to all who connect to the MQ network including students.

Changes from Previous Offering

The previous offering of this unit used videos provided via iLearn instead of lectures. In this offering, weekly lectures have been reinstated due to student feedback indicating this would be beneficial. In addition, more information about assessment tasks will be given earlier in the semester to allow student to better plan their time.

We value student feedback to be able to continually improve the way we offer our units. As such, we encourage students to provide constructive feedback via student surveys, to the teaching staff directly, or via the FSE Student Experience & Feedback link in the iLearn page.

Unit information based on version 2026.04 of the [Handbook](#)